Experiment 18
Speed of sound in air

Victoria Torres Ochando and Martina Lodos Falcón
Group A3-6

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Abstract

This experiment aimed to measure the speed of sound in air using a resonance tube partially filled with water and a tuning fork of known frequency (440Hz, 480Hz and 512Hz). By identifying the positions where resonance occurred (indicated by a noticeable increase in sound intensity) we were able to determine the length of the air column at resonance. Using these measurements, we calculated the wavelength and, with its value, the speed of sound. This experiment highlighted how acoustic resonance can be used as a reliable method for determining key physical properties of gases.

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1 Introduction

1.1 Contextualization

Before the discovery of the fact that sound traveled in waves, for years many people were bewildered and curious about how we were able to hear what we hear. This all changed when some of the most notable scientists discovered how sound travels.
One of the first people to claim that sound traveled in waves was Greek philosopher, Aristotle.

Aristotle claimed that the quality of sound will remain unchanged and travel as far as the waves reach [1].

After many breakthroughs by other researchers over time, in the 17th and 18th centuries, scientists like Isaac Newton and Pierre-Simon Laplace developed theoretical frameworks to describe the speed of sound in air, introducing concepts such as adiabatic processes in gas dynamics [2].

This can be observed in his Newton's seminal 1687 work Philosophiae Naturalis Principia Mathematica, that established the first theoretical framework for sound propagation in fluids. His analysis demonstrated that the acoustic properties of a fluid were fundamentally dependent on its elasticity and density. Newton derived what became the first mathematical expression for the speed of sound, though his calculation showed a 16% deviation from experimental values.

From then on, modern physics recognized that sound waves create rapid pressure variations where temperature changes cannot equilibrate, an adiabatic process rather than the isothermal assumption Newton employed. The correction of this fundamental oversight was later achieved by Pierre-Simon Laplace in the early 19th century, who properly accounted for the thermal dynamics of compression waves.

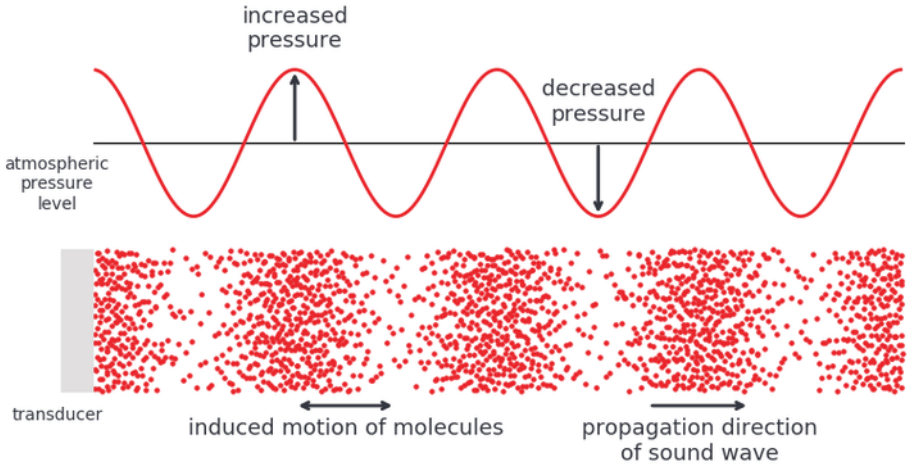


Figure 1: Properties of the propagation of a sound wave through air.

1.2 Theorical principles

1.2.1 Speed of sound in air

Sound is a longitudinal mechanical wave that propagates through a medium via pressure and density fluctuations. Its speed v in air relate to its wavelength λ and frequency f by

$$v = \lambda f \tag{1}$$

As this type of wave can propagate in different material media (solids, liquids, and gases), we particularize the wave velocity for a propagation trough gas as

$$v = \sqrt{\frac{\beta}{\rho}} \tag{2}$$

where ρ represents the density of the gas and β the compressibility of the medium, which is defined by $\beta = \frac{\partial P}{\partial V} V$, with P and V as the pressure and volume of the gas, respectively.

Given the speed of propagation of this wave, we can consider this to be an adiabatic process. Thus, for an ideal gas in this situation we relate the following:

$$PV^{\gamma} = C \tag{3}$$

where γ, the adiabatic coefficient, can be calculated as $\gamma = \frac{C_p}{C_v}$, and C is a constant. That way, resorting to the expression of β, for an adiabatic process, we obtain

$$\beta = \gamma P \tag{4}$$

On the other hand, assuming ideal gas conditions where

$$PV = nRT \tag{5}$$

and expressing density in terms of the number of moles n and the molecular mass of the gas M,

$$\rho = \frac{m}{V} = \frac{nM}{V} \tag{6}$$

we can susbtitute the equation 4 into equation 2 and find the netx expression for v:

$$v = \sqrt{\frac{\gamma RT}{M}} \tag{7}$$

under the assumption that γ is the adiabatic coefficient (approximately 1.4 for air), R suppresses the universal gas constant (8.314 J/mol·K in SI units), T is the absolute temperature in Kelvins and M relate to the molecular mass of the gas ($\approx$ 28.9 g/mol in the case of air).
Finally, for an ideal gas, the speed of sound depends only on temperature (T), since γ, M, and R are constants. Given the speed v at temperature T, the speed v_0 at another temperature (for example 0°C) can be found by applying the next expression twice and dividing the results term by term:

$$v = v_0\sqrt{\frac{T}{T_0}} \tag{8}$$

1.2.2 Resonance

Resonance is a fundamental wave phenomenon that occurs when a system is driven at its natural frequency, leading to maximum energy transfer and amplitude amplification. In acoustics, resonance manifests when sound waves constructively interfere within a bounded medium, creating standing waves with distinct nodes (points of minimal displacement) and antinodes (points of maximal displacement)[3].

For a tube closed at one end and open at the other, resonance occurs when the air column length (L) matches an odd multiple of a quarter-wavelength ($\gamma/4$):

$$L = (2n-1)\frac{\lambda}{4} \tag{9}$$
$$(n = 1, 2, 3, 4...)$$

However, in practice there is a factor to take into account: the position of the first node does not coincide exactly with the closed physical end of the tube, it differs a distance e outside the tube. In consequence, we must correct the expression:

$$L_1 + e = \frac{\lambda}{4}, \tag{10}$$

$$L_2 + e = \frac{3\lambda}{4},$$

and, from these equations we get:

$$\lambda = 2(L_1 - L_2), \tag{11}$$

$$e = \frac{L_2 - 3L_1}{2}. \tag{12}$$

With that, once calculated the wavelength λ and frequency f, the speed of sound can be determined by applying relation 1.

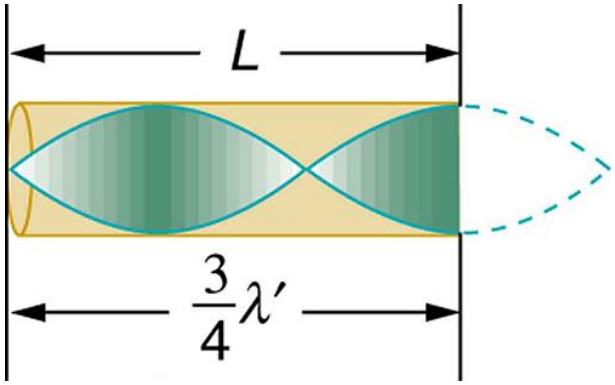


Figure 2: Resonance for a tube closed at one end. This has maximum air displacements at the open end, and none at the closed end.

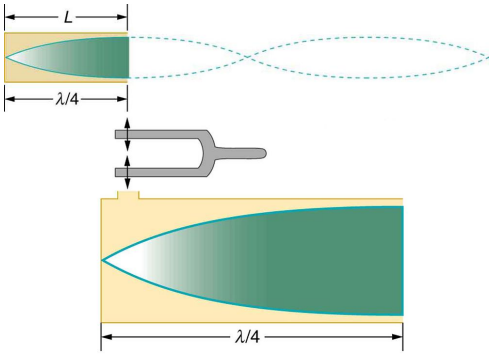


Figure 3: The standing wave is created in the tube by the vibration generated by the tuning fork introduced near its closed end.

2 Materials and Methods

2.1 Materials

For this practice, the following materials were used.

- **Resonance tube**. A vertical tube 1 meters with a internal diameter 3 cm, equipped with a graph paper background (1 mm resolution) for precise measurements.
- **Tuning forks**. Three forks with frequencies of 440 Hz, 480 Hz, and 512 Hz.
- **Mercury barometer**. For measuring atmospheric pressure (resolution: 0.05 mmHg).
- **Thermometer**. Glass tube type with a resolution of $0.5C$.
- **Rubber hammer**. To excite the tuning forks.

2.2 Methods

1. First of all, we will position the 440 Hz tuning fork vertically above the resonance tube opening, filling then both the reservoir and the resonance tube with water to near the edge of the resonance tube.
2. After striking the tuning fork with the rubber hammer, we will gradually lower the water level until observing an intensification of the sound.
3. Upon identifying the approximate resonance point, we will record the exact distance L_1 (water surface to tube edge) and repeat this measure three times according to error theory. Then, L_2 will be determined by following the same process. With these values, using equations 11 and 12, we will be able to find wavelength λ and the correction e.
4. Knowing the value of λ and the frequency f of the tuning fork, it will be sufficient to apply expression 1 to find the speed of sound in air.
5. To continue or calculus, we will need to record the barometric pressure P (with mercury temperature correction from ASTM tables) and the ambient temperature T. In addition, applying equation 8, we will calculate sound speed at $0C$.

6. To this point, we will calculate the adiabatic coefficient (γ) and the modulus of compressibility (β) of air with 4 and 7. Finally, the molar heats at constant pressure and volume for air will be find with the determined value of γ and

$$\gamma = \frac{c_p}{c_v} \tag{13}$$

$$c_p - c_v = R$$

This process will be repeated for the 480Hz and the 512Hz tuning forks [4].

3 Results and discussion

In order to present the results obtained in a more visual way, we will divide this section into three parts, one for each tuning fork used.

3.1 440 Hz tuning fork

The data collected using the first tuning fork are presented in tables 1 and 2.

$v(m/s)$	$v_0(m/s)$	$\lambda(m)$	$e(m)$
$343,2 \pm 1,4$	$329,7 \pm 1,4$	$0,7800 \pm 0,0031$	$0,0147 \pm 0,0015$

Table 1: Speed of sound in air at a temperature of $23 \pm 0,14°C$, v, and at $0°C$, v_0; the wavelength λ in this conditions; and e, the position of the first maximum.

γ	$\beta(Pa)$	$c_v(J/mol \cdot K)$	$c_p(J/mol \cdot K)$
$1,383 \pm 0,011$	140392 ± 1100	$21,70 \pm 0,62$	$30,01 \pm 0,89$

Table 2: γ is the adiabatic coefficient; β, the compressibility modulus; and c_v and c_p are the specific heats given a constant volume and pressure respectively.

The value obtained for the speed of sound at a temperature of $23 \pm 0,14°C$ is $343,20 \pm 0,64$ m/s, quite close to the one we were expecting (343 m/s at $20°C$, according to [5]). Similarly, the calculated speed at 0°C agrees with its real value (331 m/s, [5]).

The derived adiabatic coefficient, $\gamma = 1,383 \pm 0,011$, aligns closely with the theoretical value for air $\gamma \approx 1,40$, as stated in [6]. In addition, the compressibility modulus β has a value quite close to its referential value, $1,41 \cdot 10^5 Pa$ ([5]) *vs.* the one we obtained, $140392 \pm 1100 Pa$.

However, the specific heat values c_p and c_v are marginally different than the standard values ($c_p \approx 29,1 J/molK$ and $c_v \approx 20,8 J/molK$) [5].

3.2 480 Hz tuning fork

Tables 3 and 4 display the measurements recorded using the 480 Hz tuning fork.

$v(m/s)$	$v_0(m/s)$	$\lambda(m)$	$e(m)$
$345,3 \pm 1,1$	$331,59 \pm 1,05$	$0,7193 \pm 0,0022$	$0,0168 \pm 0,0011$

Table 3: Speed of sound in air at a temperature of $23 \pm 0,14°C$, v, and at $0°C$, v_0; the wavelength λ in this conditions; and e, the position of the first maximum.

γ	$\beta(Pa)$	$c_v(J/mol \cdot K)$	$c_p(J/mol \cdot K)$
$1,4000 \pm 0,0088$	142100 ± 900	$20,78 \pm 0,46$	$29,10 \pm 0,67$

Table 4: γ is the adiabatic coefficient; β, the compressibility modulus; and c_v and c_p are the specific heats given a constant volume and pressure respectively.

With this tuning fork the measured speed of sound at room temperature is a bit bigger than its theoretical value, $345,3 \pm 1,1$ versus 343 m/s. The value calculated for $0°C$, $331,59 \pm 1,05$ m/s, shows strong agreement with the standard reference, i.e., 331 m/s [5].

The adiabatic coefficient obtained from the experiment is in good alignment with the theoretical value for air. Likewise, the compressibility modulus is consistent with the commonly accepted standard, showing minimal deviation from it.

In addition, in this experiment the values found for the specific heats at constant volume and pressure, $20,78 \pm 0,46$ and $29,10 \pm 0,67$ respectively, are much more closer to reality [5] than with the other tuning fork.

3.3 512 Hz tuning fork

The results obtained with the 512 Hz tuning fork are shown in tables 5 and 6.

$v(m/s)$	$v_0(m/s)$	$\lambda(m)$	$e(m)$
$345,8 \pm 1,2$	$332,1 \pm 1,1$	$0,6753 \pm 0,0023$	$0,0185 \pm 0,0014$

Table 5: Speed of sound in air at a temperature of $23 \pm 0,14°C$, v, and at $0°C$, v_0; the wavelength λ in this conditions; and e, the position of the first maximum.

γ	$\beta(Pa)$	$c_v(J/mol \cdot K)$	$c_p(J/mol \cdot K)$
$1,4040 \pm 0,0094$	142504 ± 960	$20,58 \pm 0,48$	$28,89 \pm 0,70$

Table 6: γ is the adiabatic coefficient; β, the compressibility modulus; and c_v and c_p are the specific heats given a constant volume and pressure respectively.

In this part of the experience the determined speed of sound v is even bigger than in the second experiment. This also happens with v_0, the value extrapolated for $0°C$.

The calculated adiabatic index $(1,4040 \pm 0,0094)$ is very close to the theoretical benchmark for air [6], indicating the accuracy of the measurement. In the same way, the compressibility modulus obtained, $142504 \pm 960Pa$, is in line with the standard reference value [5]. As in the experiment done with the 480 Hz tuning fork, the specific heats at constant pressure and volume exhibit a satisfactory value compared to their reference number [5].

Regarding the wavelength λ, we can say that the values obtained decrease with the increasing frequency used, as we expected since λ is inversely proportional to f.

4 Conclusions

This experiment successfully demonstrated a practical and accessible method for measuring the speed of sound in air using acoustic resonance in a water-filled tube. Respecting our results, we can say that they were consistent with the theoretical values expected at the measured room temperature, confirming the reliability and validity of the approach.

However, some small differences between the experimental and theoretical results were observed, but we can attribute them to measurement uncertainties, such as the difficulty in finding

the exact location of maximum resonance -which was more challenging with higher frequencies because the distance was smaller, explaining the increasing error of v during the experiment- or minor temperature fluctuations during the experiment.

5 Appendixes

5.1 Calculation of Uncertainties

As we know, type A uncertainty, $u_A(x)$, which is associated to human error, can be calculated with the following equation:

$$u_A(x) = \frac{\sigma(x)}{\sqrt{N}}, \tag{14}$$

where $\sigma(x)$ represents the standard deviation and N the number of observations. On the other hand we get type B uncertainty, $u_B(x)$, related to the measuring instrument error from:

$$u_B(x) = \frac{\delta_x}{\sqrt{12}}, \tag{15}$$

where δ_x is the resolution of the measuring instrument (δ_L=0,001 m) From them, $u_C(x)$ is expressed as:

$$u_C(x) = \sqrt{u_A^2(x) + u_B^2(x)}. \tag{16}$$

Additionally, we can calculate the expanded uncertainty, which is calculated to overestimate the error. To do so, we multiplied $u_C(x)$ by a coverage factor kp that is selected for convenience:

$$U_C(x) = u_C(x)k_p. \tag{17}$$

This factor is derived from the distribution of the t-student, using $n-1$ degrees of freedom and a fixed confidence level. In this case, consistently 95%:

$$kp = 4.303.$$

So, for $L1$ and $L2$, we get:

$$u_C(L)(m) = \sqrt{u_A^2(L) + u_B^2(L)}(m). \tag{18}$$

Regarding the error of the calculated λ, since it is an indirect measurement obtained from L_1 and L_2, we express:

$$u_C(\lambda)(m) = \sqrt{\left(\frac{\partial\lambda}{\partial L_2}\right)^2 (u_C(L_2))^2 + \left(\frac{\partial\lambda}{\partial L_1}\right)^2 (u_C(L_1))^2} \tag{19}$$

$$= \sqrt{(2)^2((u_C(L_2))^2 + (-2)^2((u_C(L_1))^2} = 2\sqrt{u_C(L_2)^2 + u_C(L_1)^2}(m).$$

To continue, speed of Sound, v, obtained indirectly from equation 1, obtains an uncertainty $u_C(v)$:

$$u_C(v)(m/s) = \sqrt{\left(\frac{\partial v}{\partial\lambda}\right)^2 (u_C(\lambda))^2 + \left(\frac{\partial v}{\partial\lambda}\right)^2 (u_C(f))^2} = f \cdot u_C(\lambda)(m/s), \tag{20}$$

(we assume $u_C(f)$ equals 0). For distance e, from equation 12, we get

$$u_C(e)(m) = \sqrt{\left(\frac{\partial e}{\partial L_2}\right)^2 (u_C(L_2))^2 + \left(\frac{\partial e}{\partial L_1}\right)^2 (u_C(L_1))^2} \tag{21}$$

$$= \sqrt{\left(\frac{1}{2}\right)^2 (u_C(L_2))^2 + \left(\frac{-3}{2}\right)^2 (u_C(L_1))^2} = \frac{1}{2}\sqrt{(u_C(L_2))^2 + 9(u_C(L_1))^2}.$$

On the other hand, γ receives an uncertainty of

$$u_C(\gamma) = \sqrt{\left(\frac{\partial \gamma}{\partial v}\right)^2 (u_C(v))^2 + \left(\frac{\partial \gamma}{\partial T}\right)^2 (u_C(T))^2} = \sqrt{\left(\frac{2M}{RT}v\right)^2 (u_C(v))^2 + \left(\frac{-v^2 M}{RT^2}\right)^2 (u_C(T))^2}. \tag{22}$$

To continue, β from equation 4, gets

$$u_C(\beta) = \sqrt{P^2(u_C(\beta))^2 + \beta^2(u_C(P))^2} \tag{23}$$

As them, for C_v and C_p, we obtain

$$u_C(c_v) = \sqrt{\left(\frac{\partial c_v}{\partial R}\right)^2 (u_C(R))^2 + \left(\frac{\partial c_v}{\partial \gamma}\right)^2 (u_C(\gamma))^2} = \left|\frac{-R}{(\gamma-1)^2}\right| u_C(\gamma) \tag{24}$$

$$= \left(\frac{R}{(\gamma-1)^2}\right) u_C(\gamma)(J/molK),$$

$$u_C(c_p) = \sqrt{\left(\frac{\partial c_p}{\partial c_v}\right)^2 (u_C(c_v))^2 + \left(\frac{\partial c_p}{\partial \gamma}\right)^2 (u_C(\gamma))^2} = \sqrt{\gamma^2(u_C(c_v))^2 + c_v^2(u_C(\gamma))^2}(J/molK). \tag{25}$$

Finally, for v_0, from 8

$$u_C(v_0) = \sqrt{\left(\frac{\partial v_0}{\partial v}\right)^2 (u_C(v))^2 + \left(\frac{\partial v_0}{\partial T}\right)^2 (u_C(T))^2} \tag{26}$$

$$= \sqrt{\left(\sqrt{\frac{T_0}{T}}\right)^2 (u_C(v))^2 + \left(\frac{-v}{2T}\sqrt{\frac{T_0}{T}}\right)^2 (u_C(T))^2}.$$

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